

BASIC TERMS IN MAGNETISM

Magnetisation Or Intensity of Magnetisation ($\vec{M}$) :

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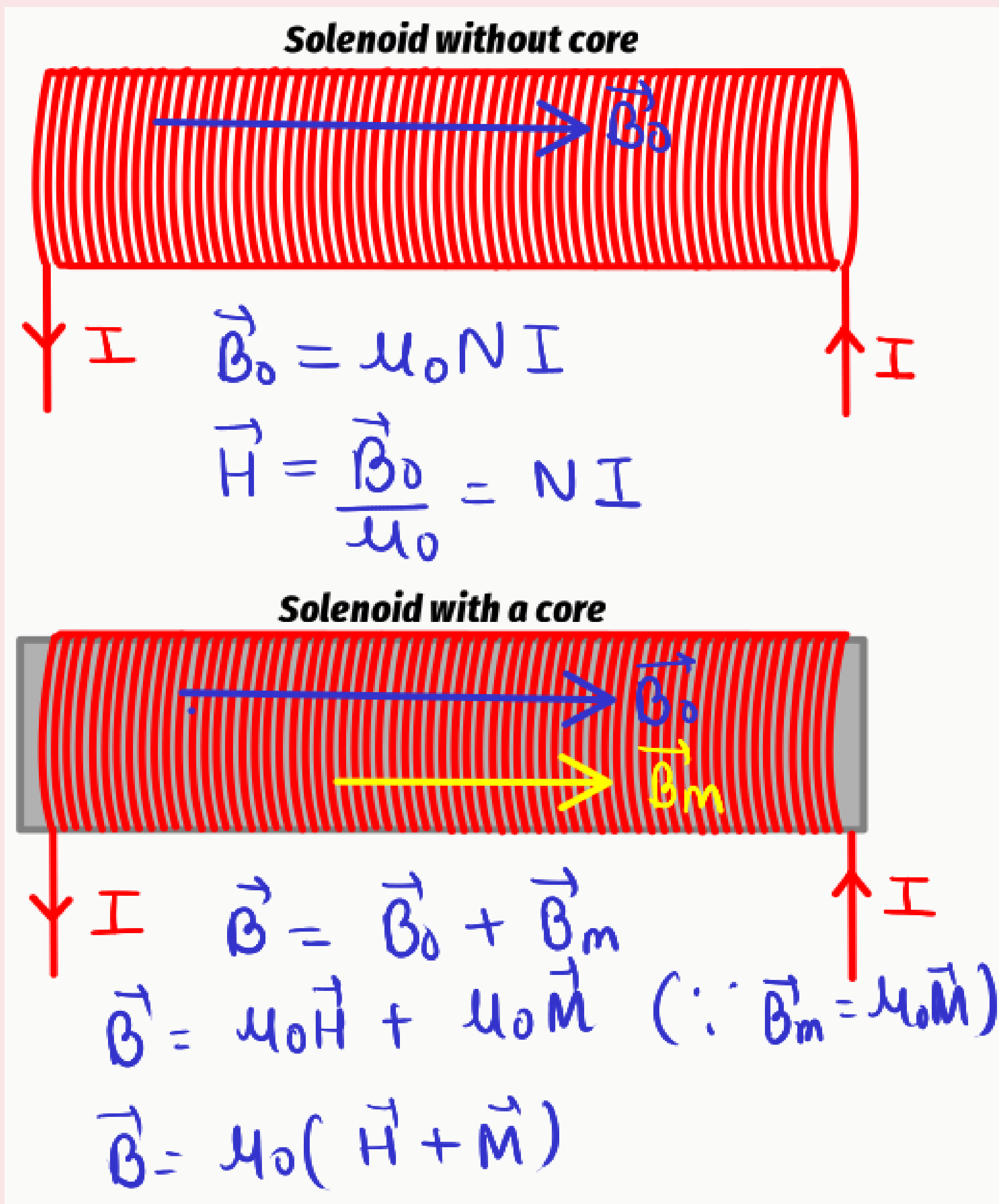
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- S.I. Unit of $\vec{M} = \frac{A \, m^2}{m^3} = A \, m^{-1}$

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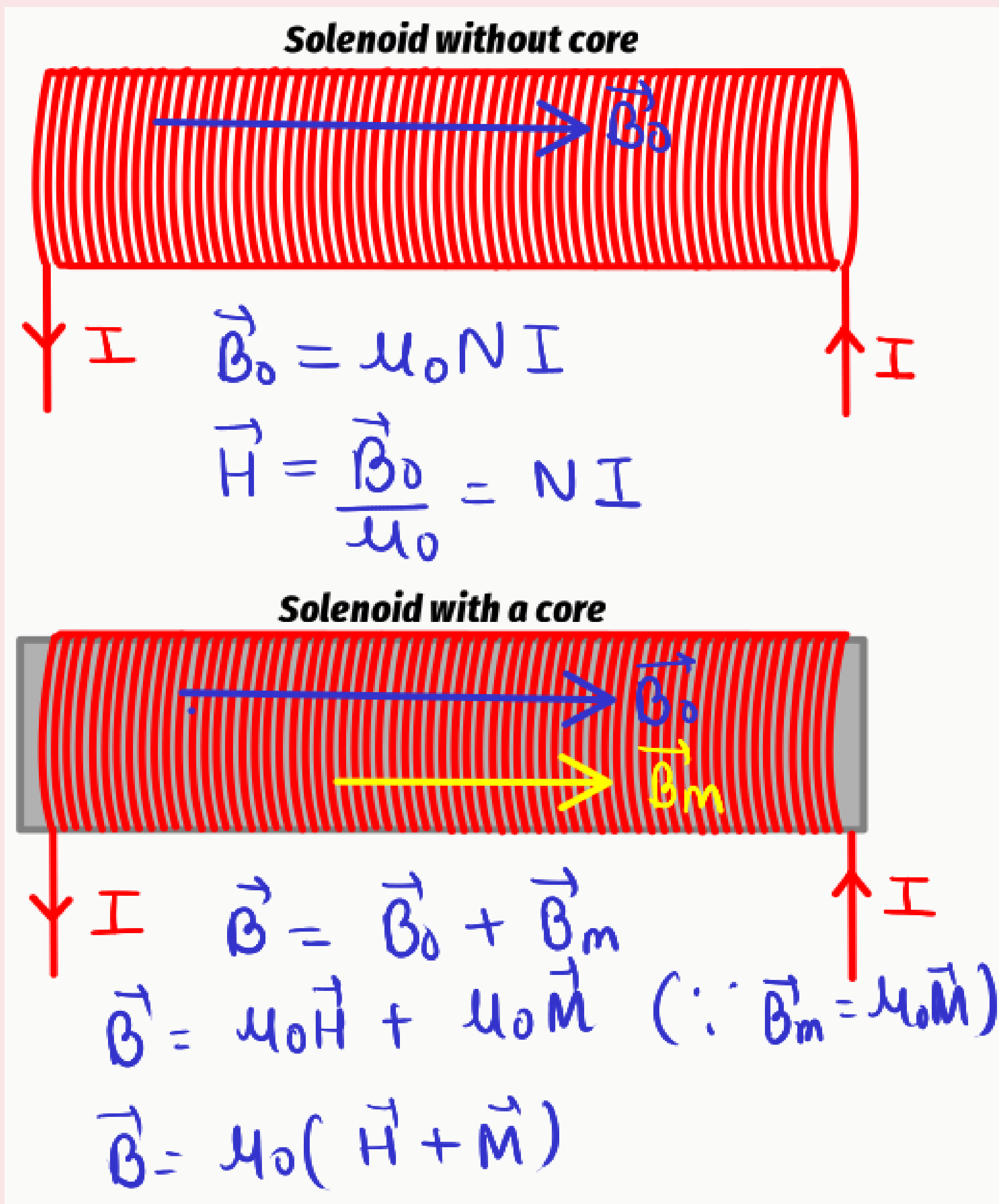
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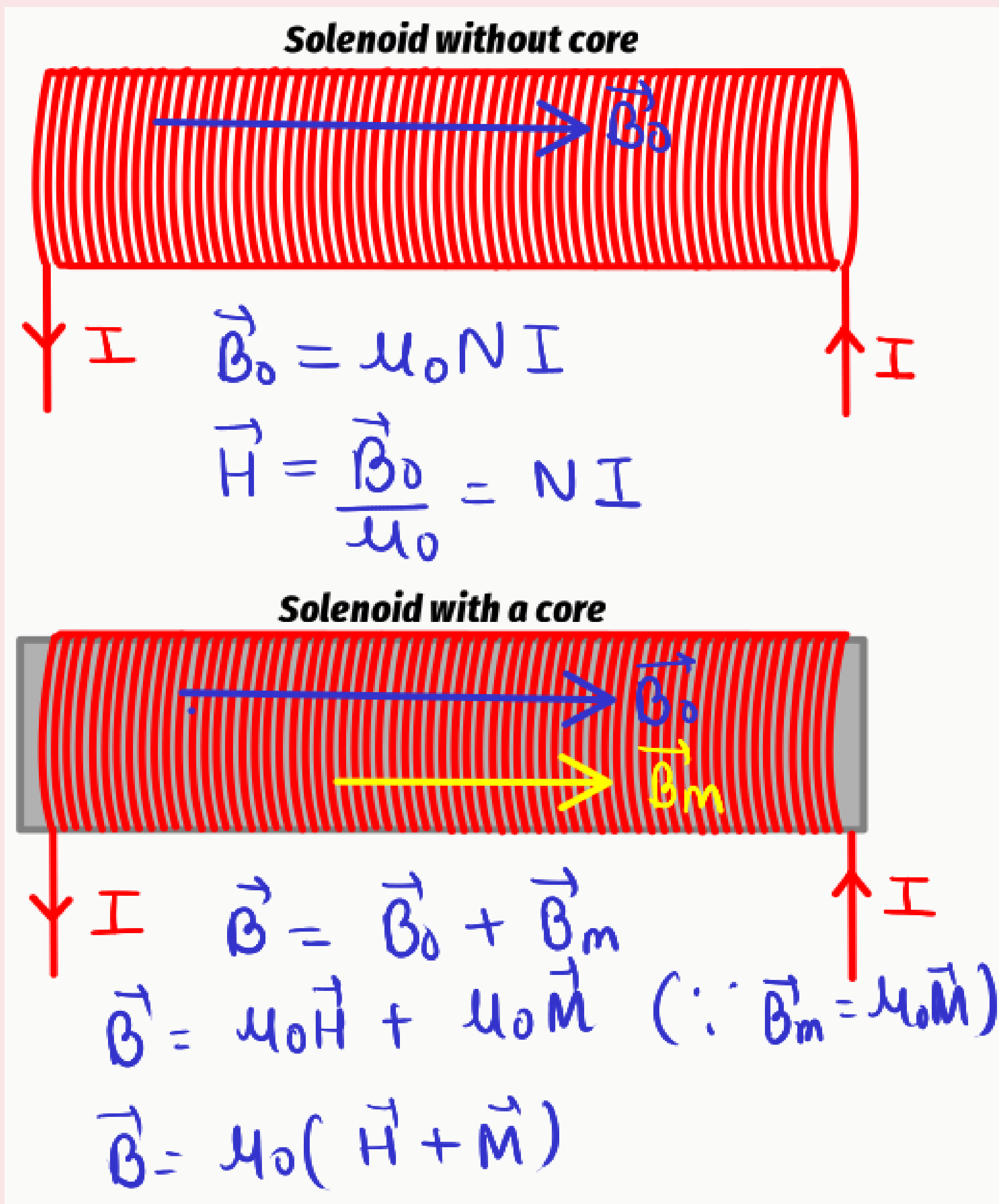
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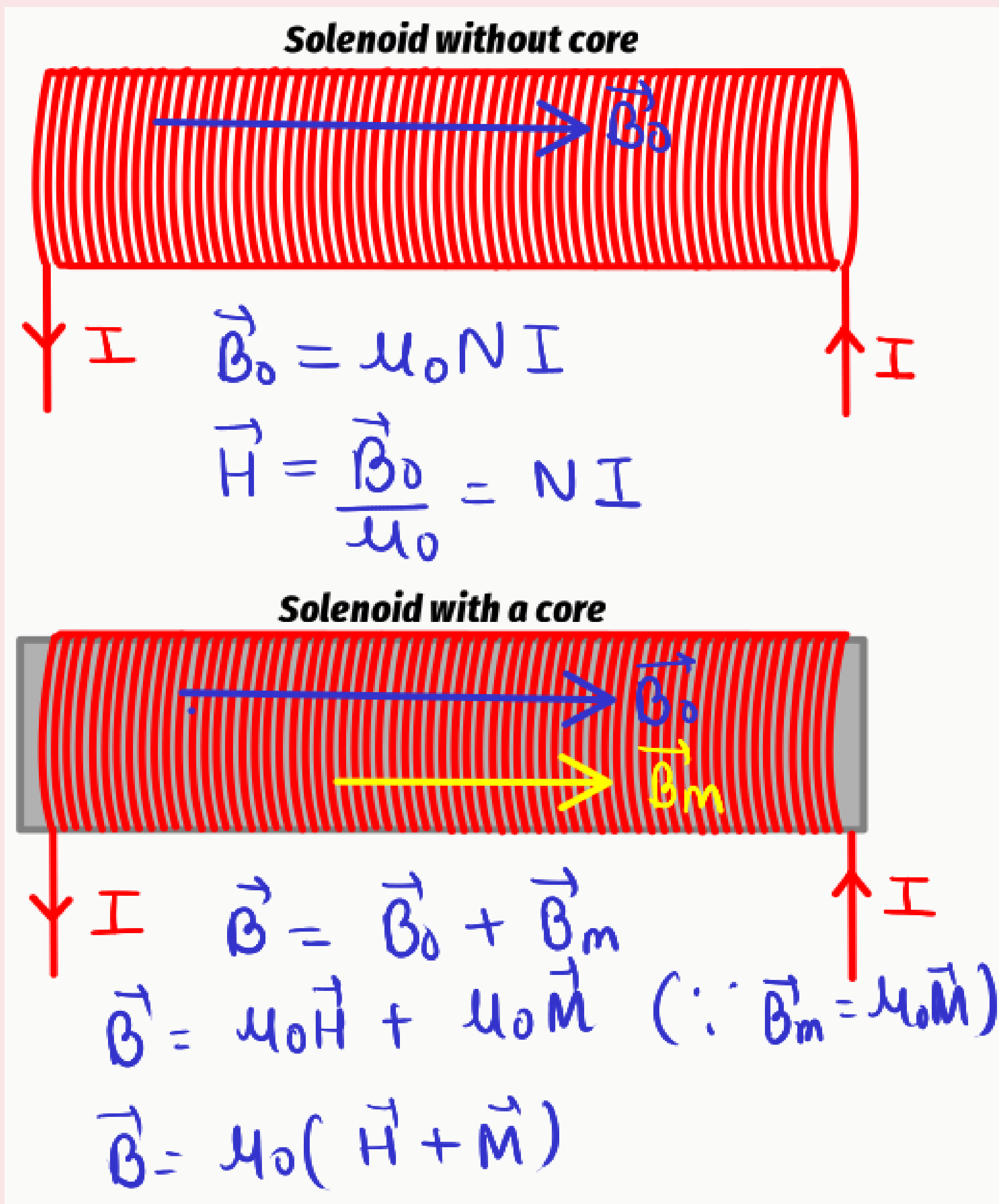


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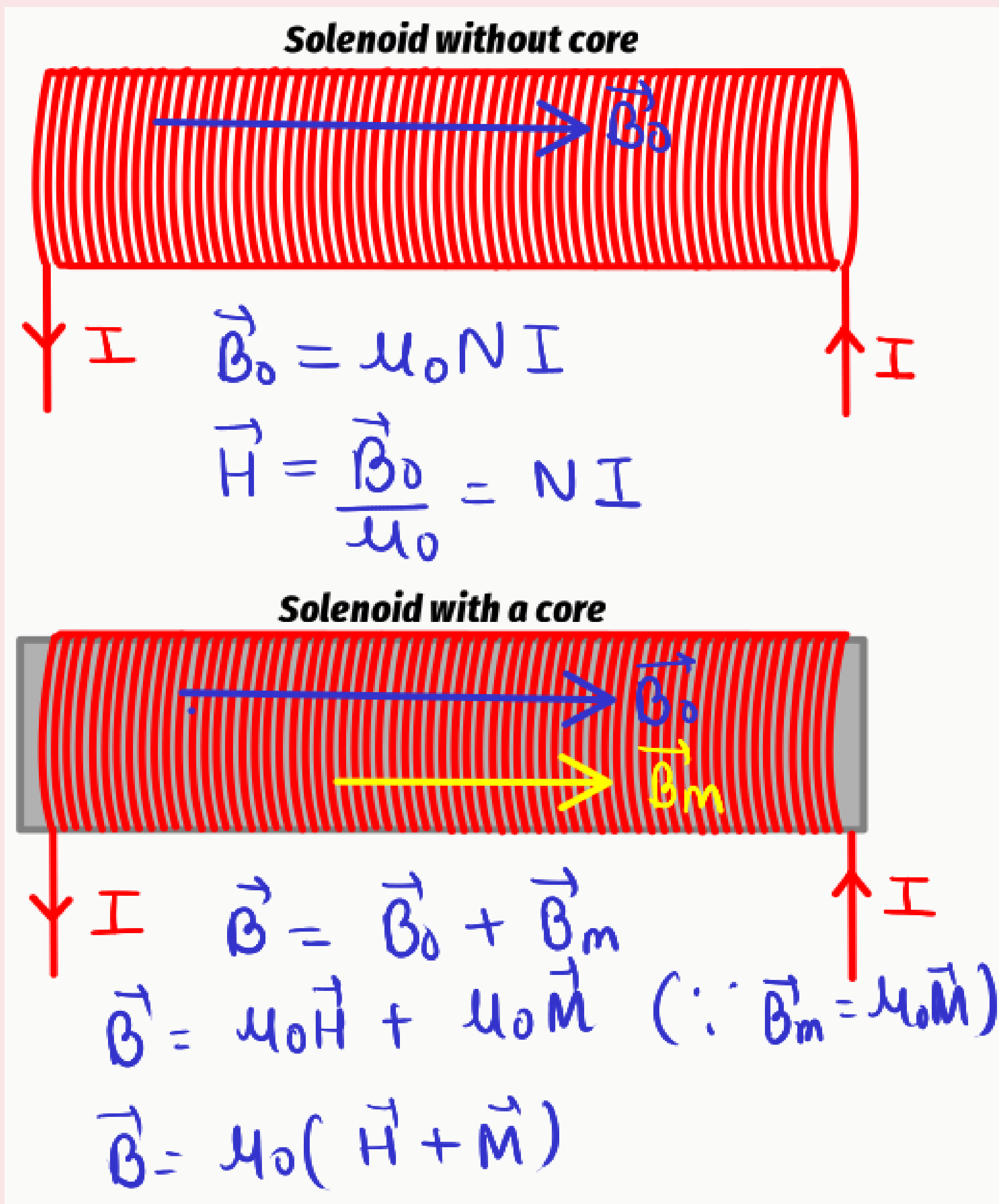
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- χ is called **magnetic susceptibility** it is dimensionless quantity. for vacuum $\chi = 0$

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Example 5.10

A solenoid has a core of a material with relative permeability 400. The windings of the solenoid are insulated from the core and carry a current of 2 A. If the number of turns is 1000 per metre, calculate (a) H , (b) B , (c) M and (d) the magnetising current I_m .

Classification of magnetic materials

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Diamagnetic	Paramagnetic	Ferromagnetic
$-1 \leq \chi < 0$ $0 \leq \mu_r < 1$ $\mu < \mu_0$	$0 < \chi < \varepsilon$ $1 < \mu_r < 1 + \varepsilon$ $\mu > \mu_0$ Here ε is a small positive number	$\chi \gg 1$ $\mu_r \gg 1$ $\mu \gg \mu_0$

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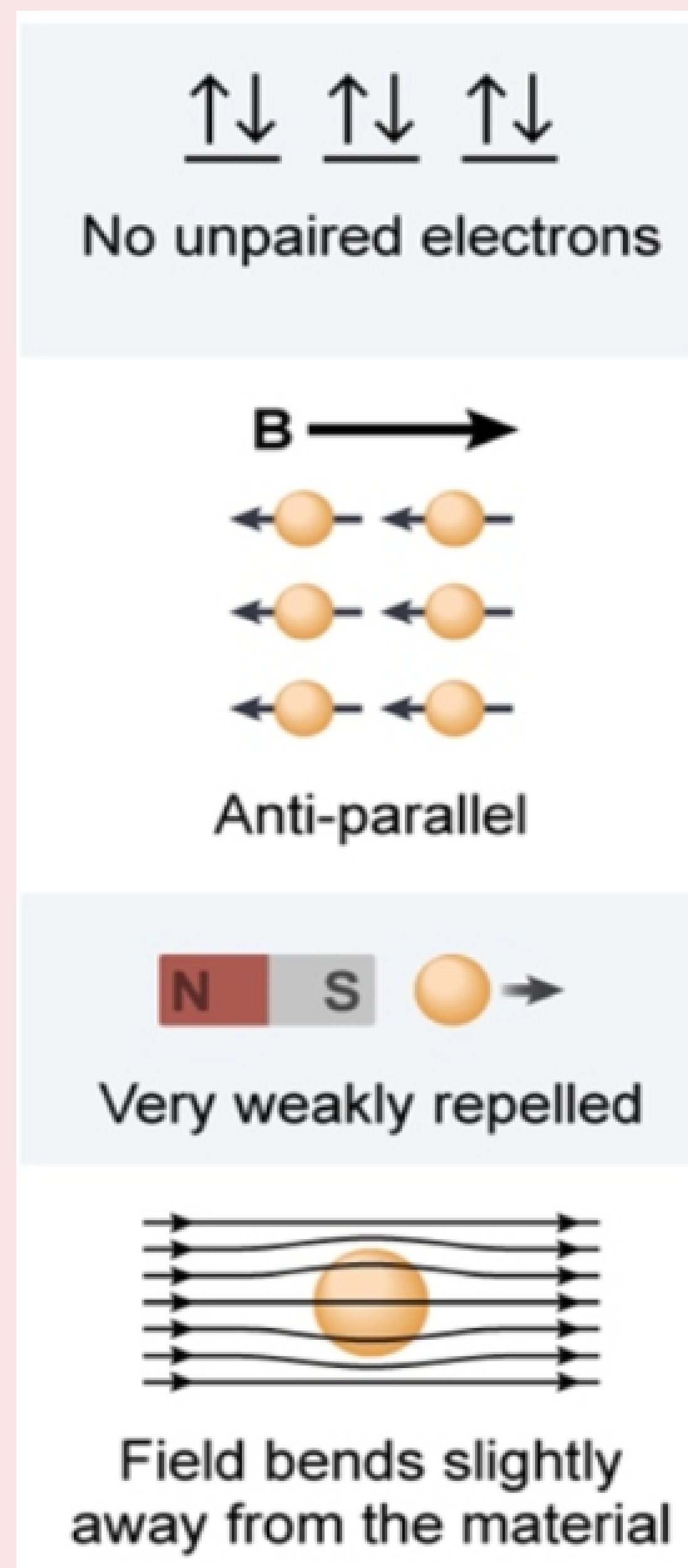
- Those substances which are weakly magnetised in a direction opposite to the applied magnetic field are called diamagnetic substances.
- **Examples:** Gold, silver lead, copper, antimony, bismuth, silicon, mercury, water, air, hydrogen, nitrogen etc..
- **Explanation :** In general, the magnetic moments of atoms are oriented such that **the vector sum of the magnetic moments of atoms is zero**. When such a diamagnetic substance is placed in an external magnetic field, the substance as a whole acquires a net magnetic moment in a direction opposite to the applied field. Atoms of diamagnetic materials do not possess any unpaired electrons.

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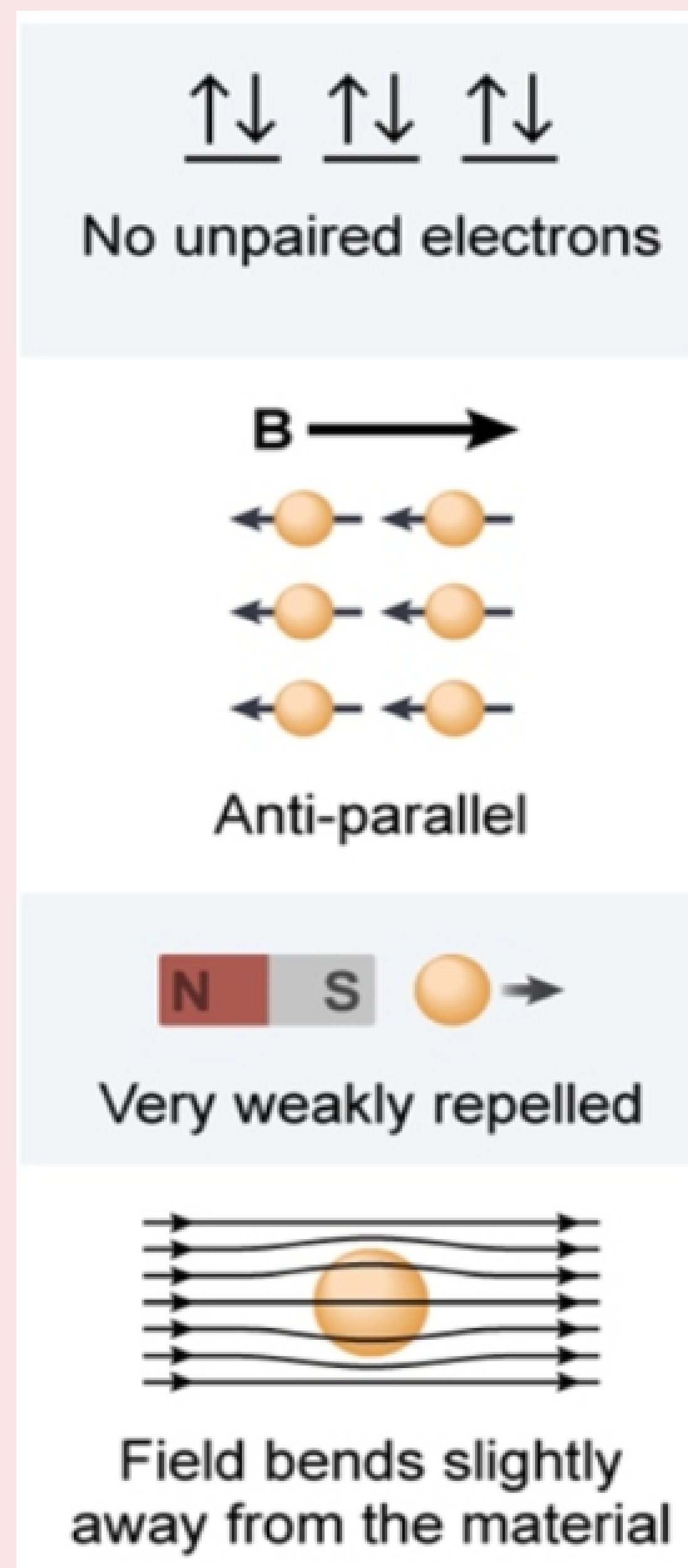


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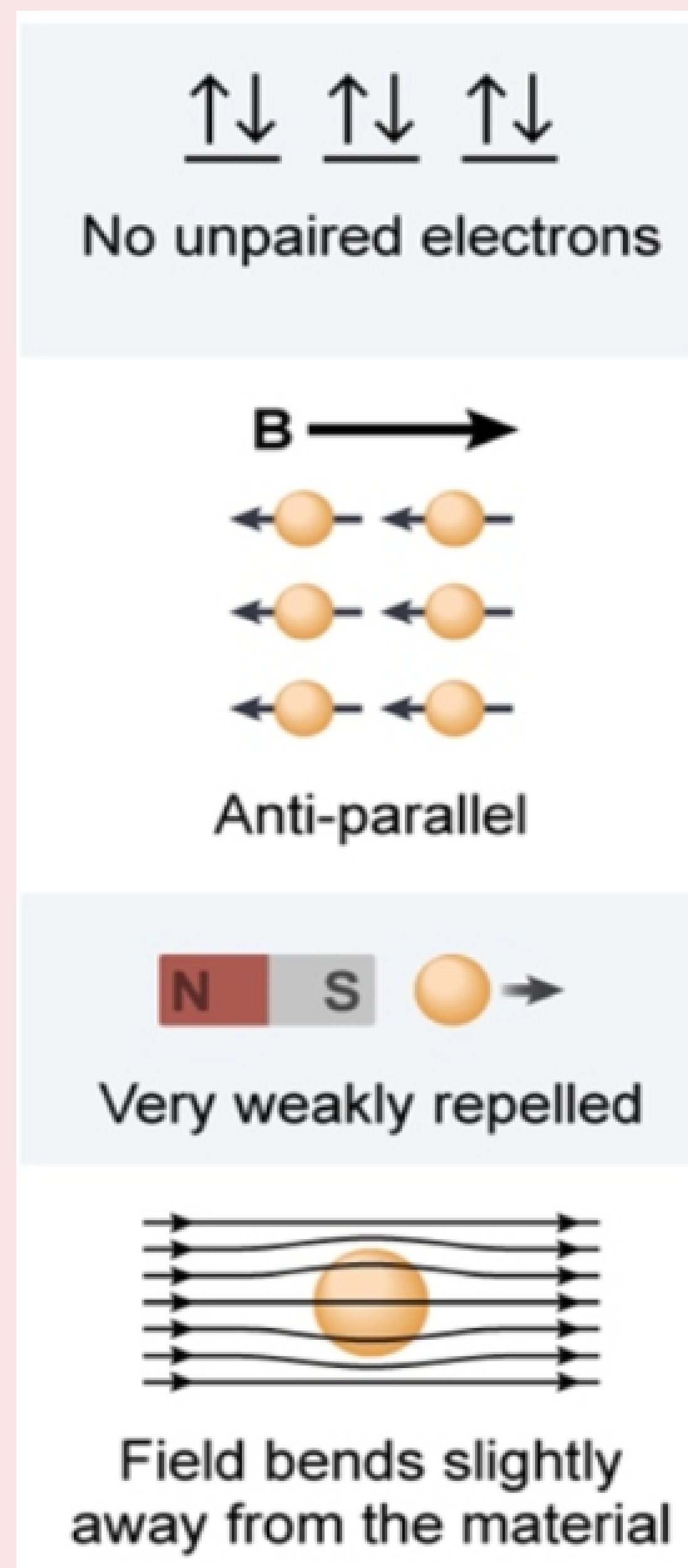


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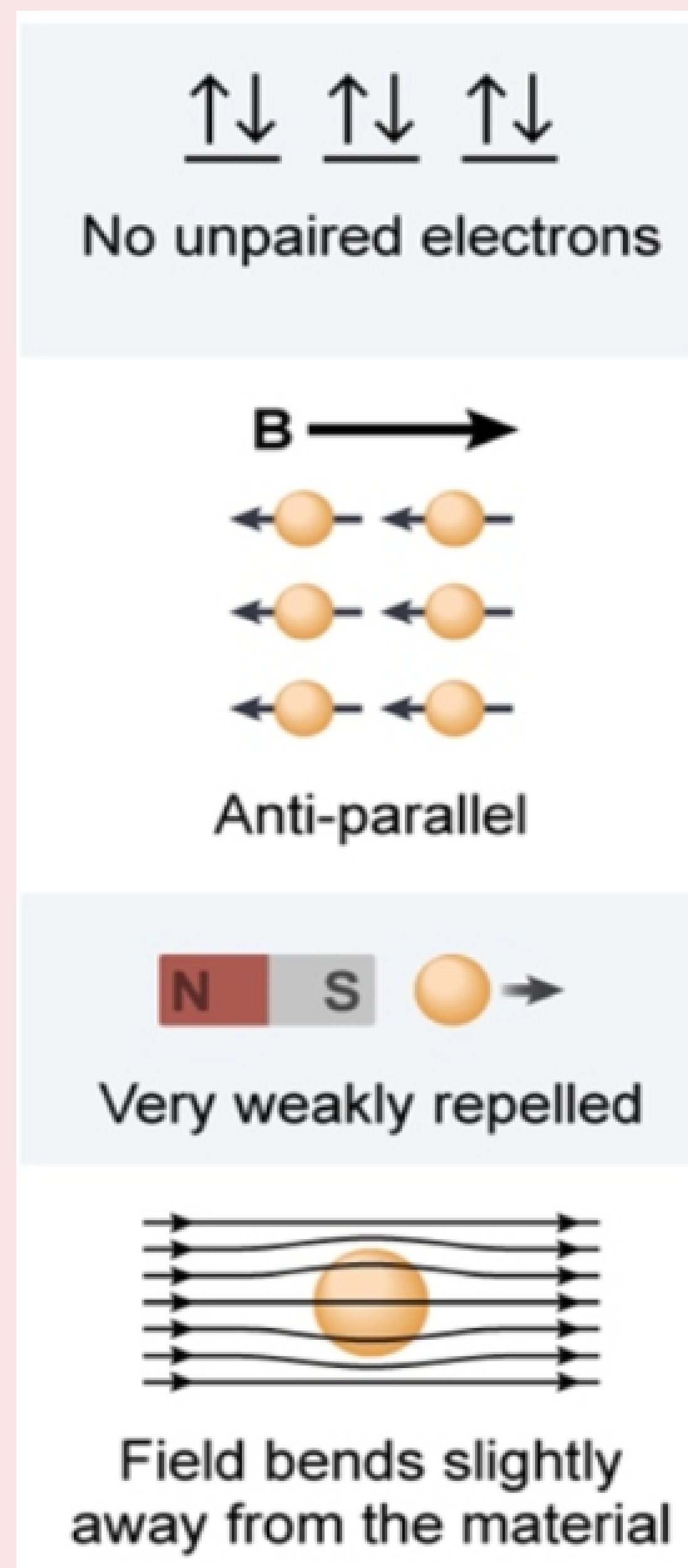


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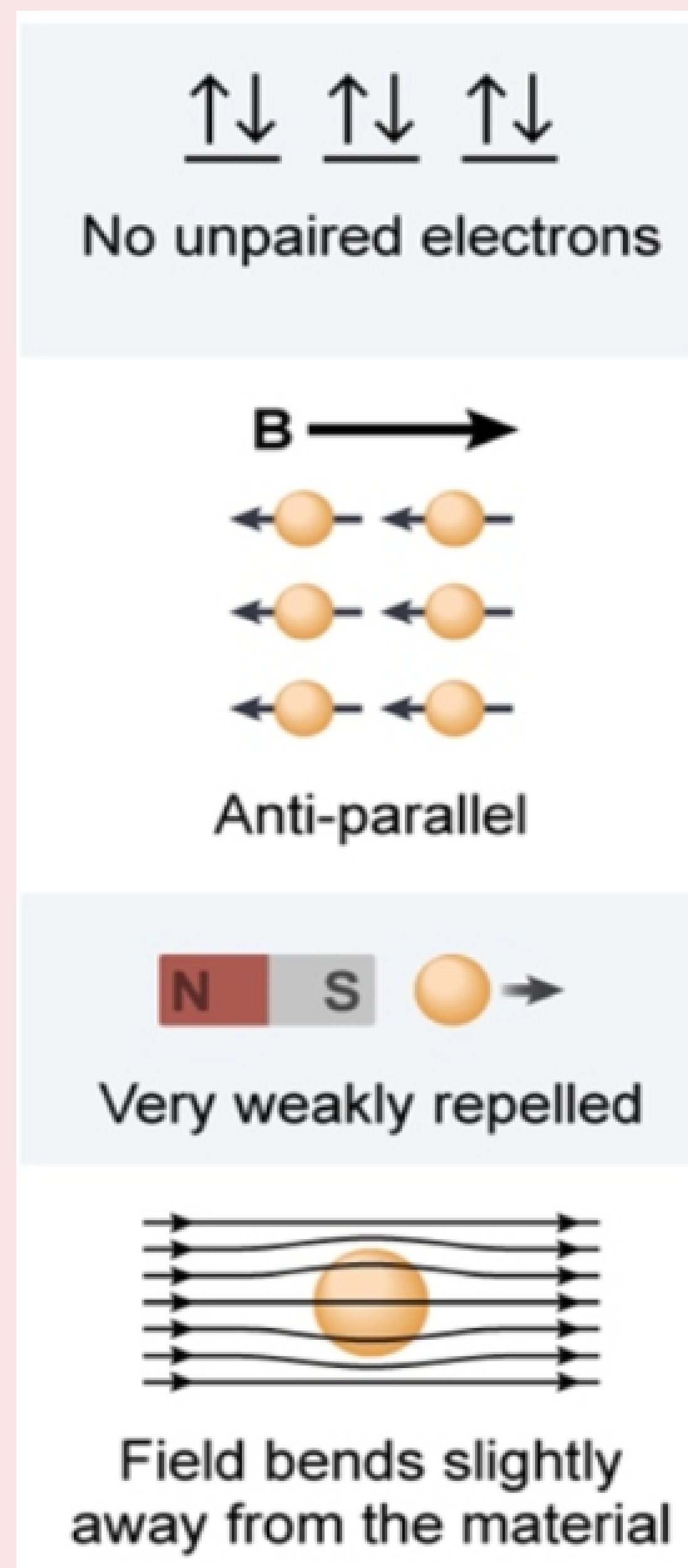


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- They **do not retain the magnetic properties** when the **external field is removed**.

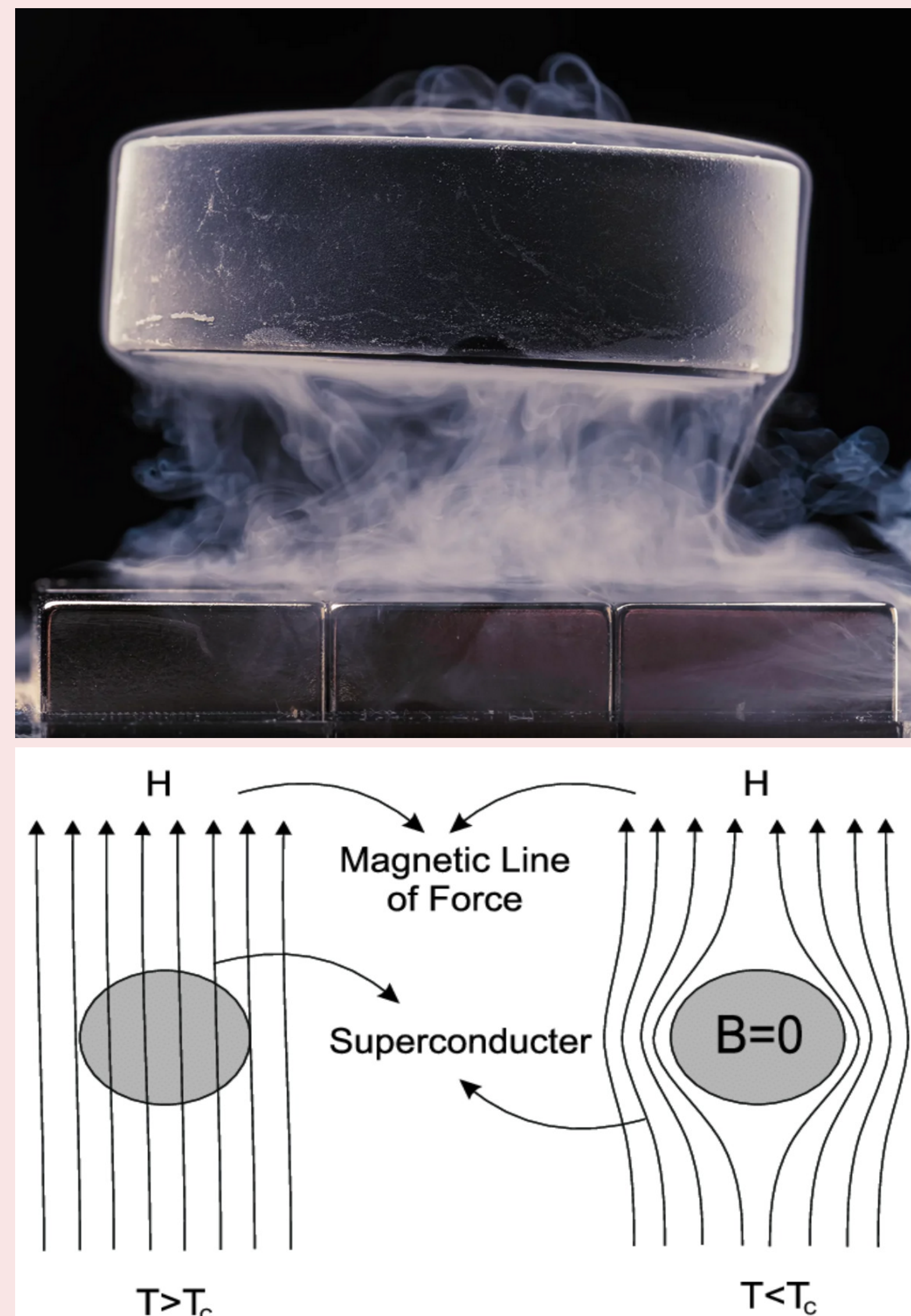


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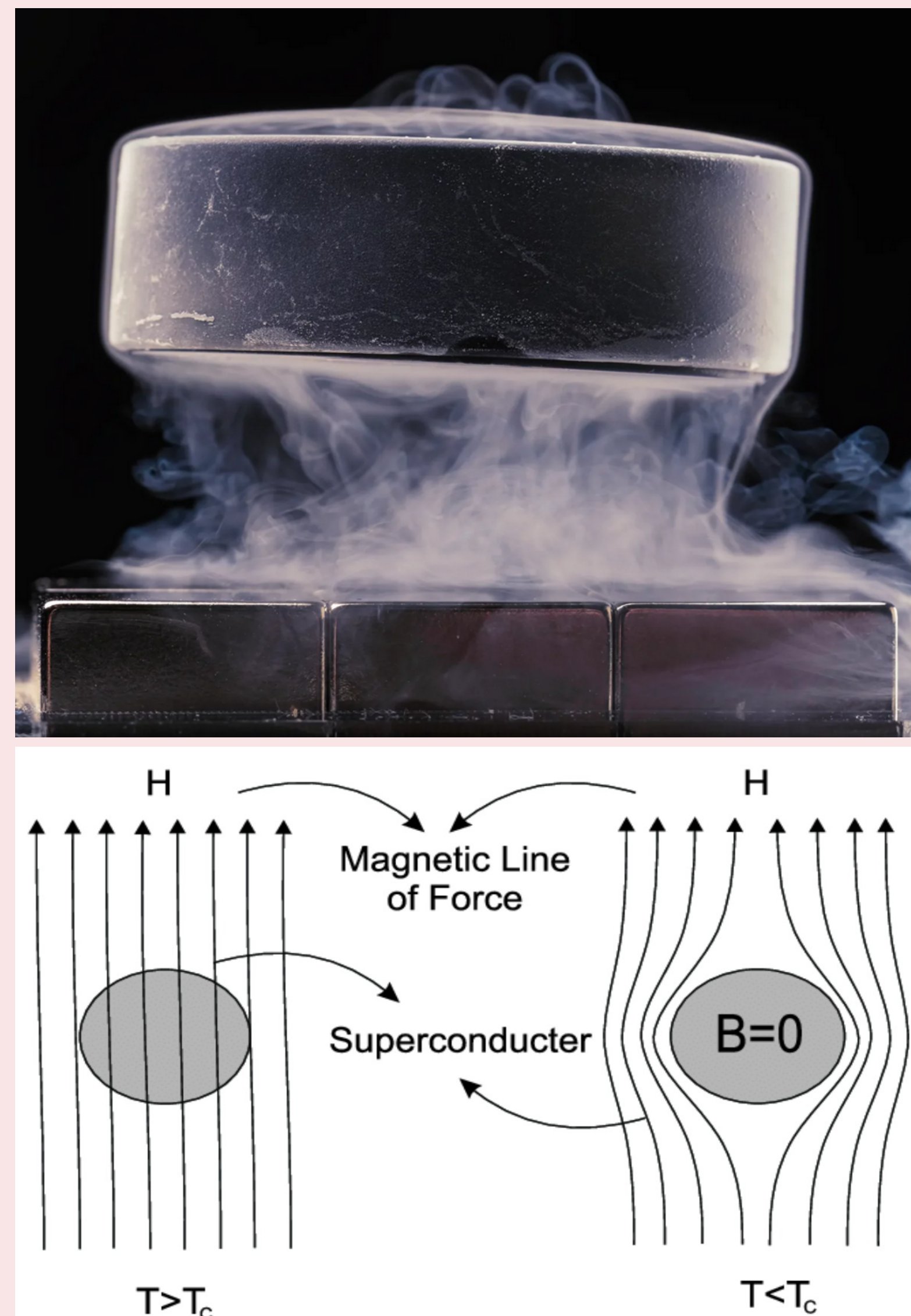


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- **Superconductors** – Superconductors are essentially diamagnetic materials with a volume **susceptibility** of $\chi = -1$.

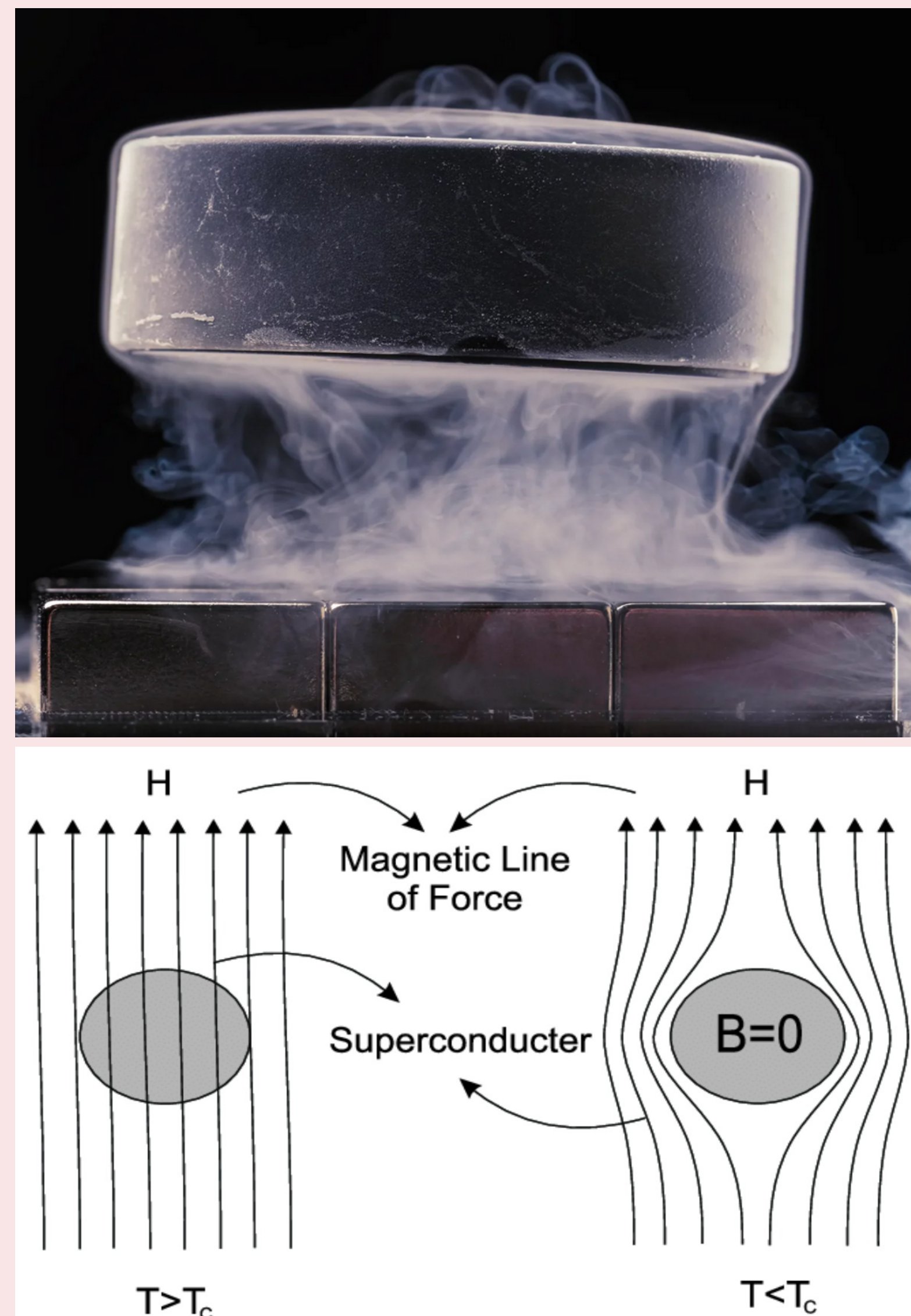


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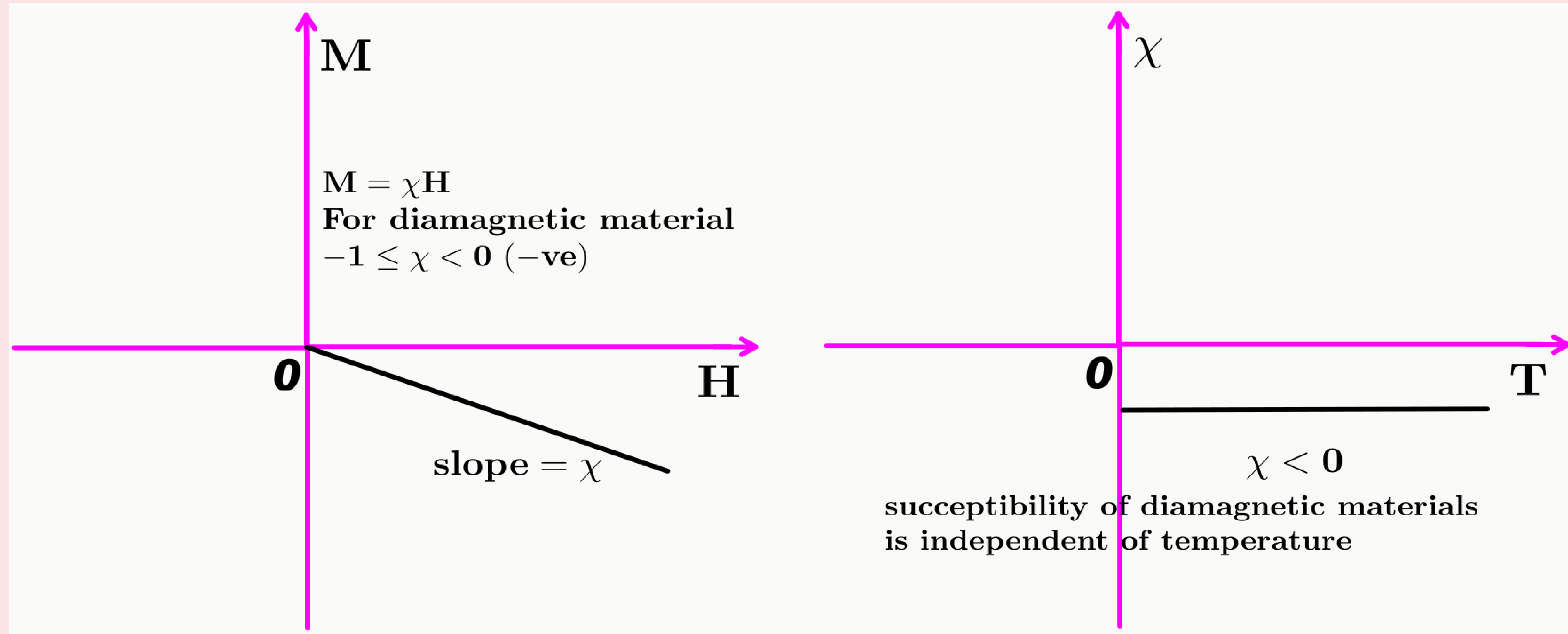
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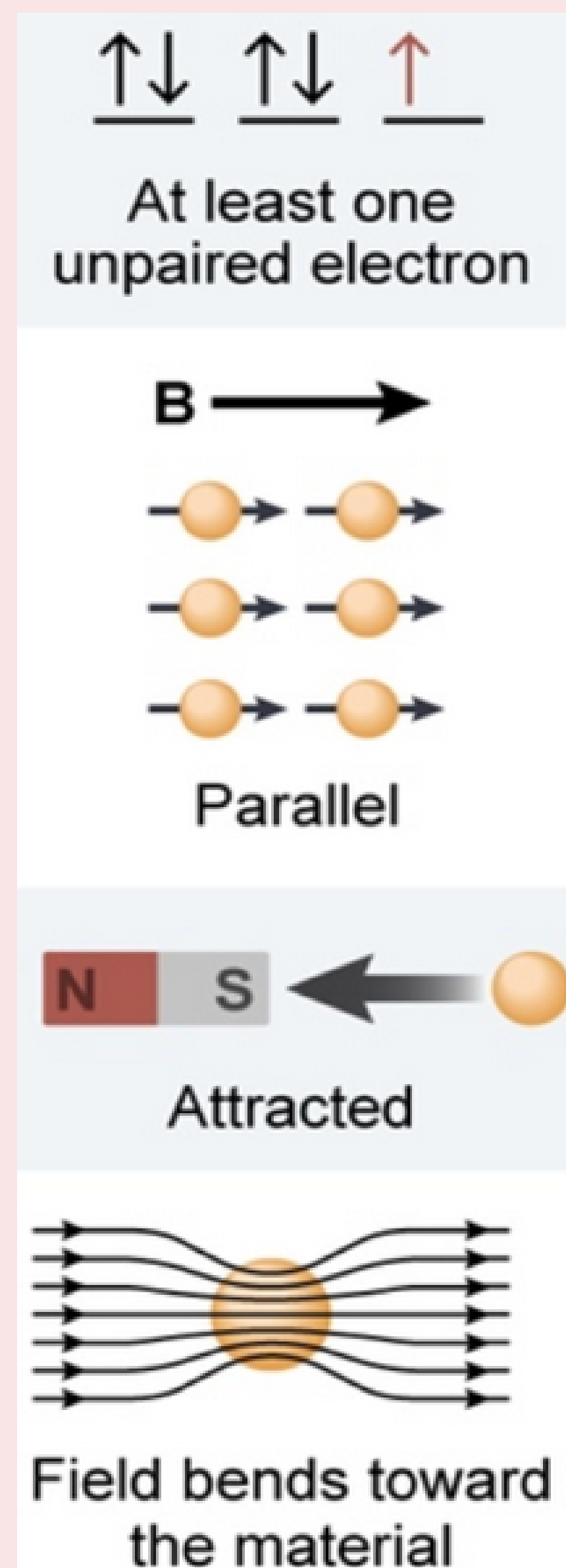
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Classification of magnetic materials

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Characteristics :

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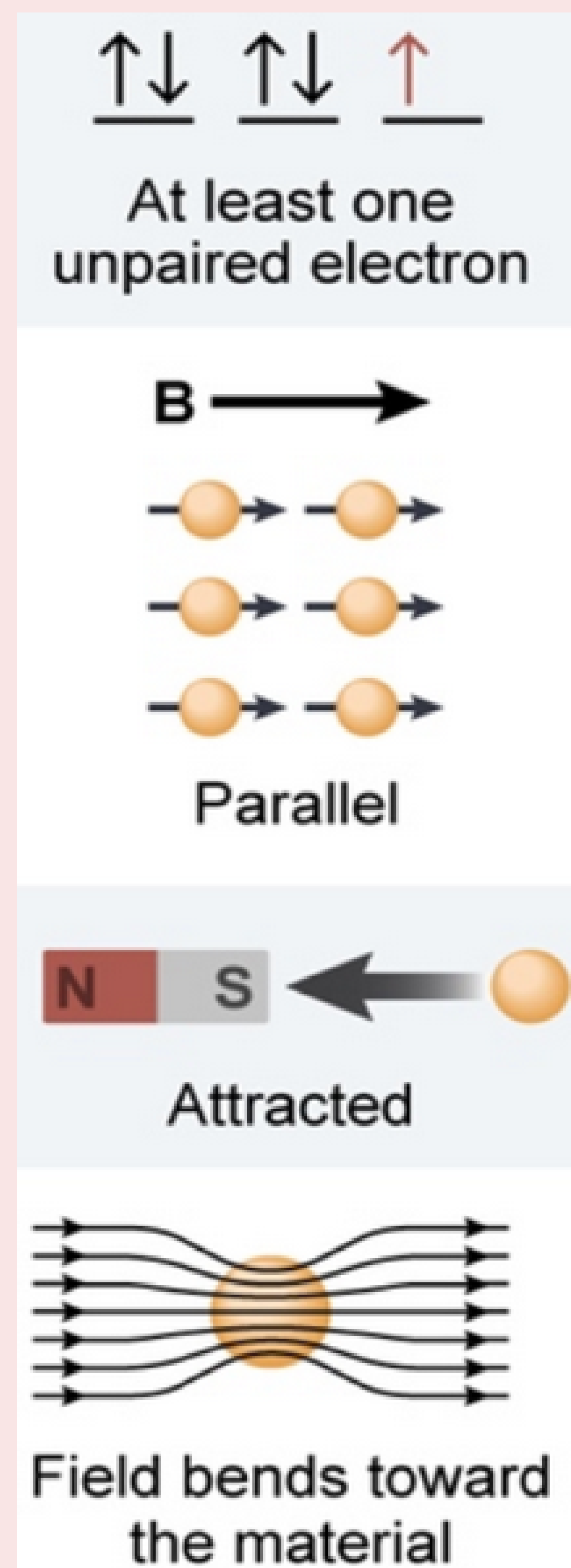


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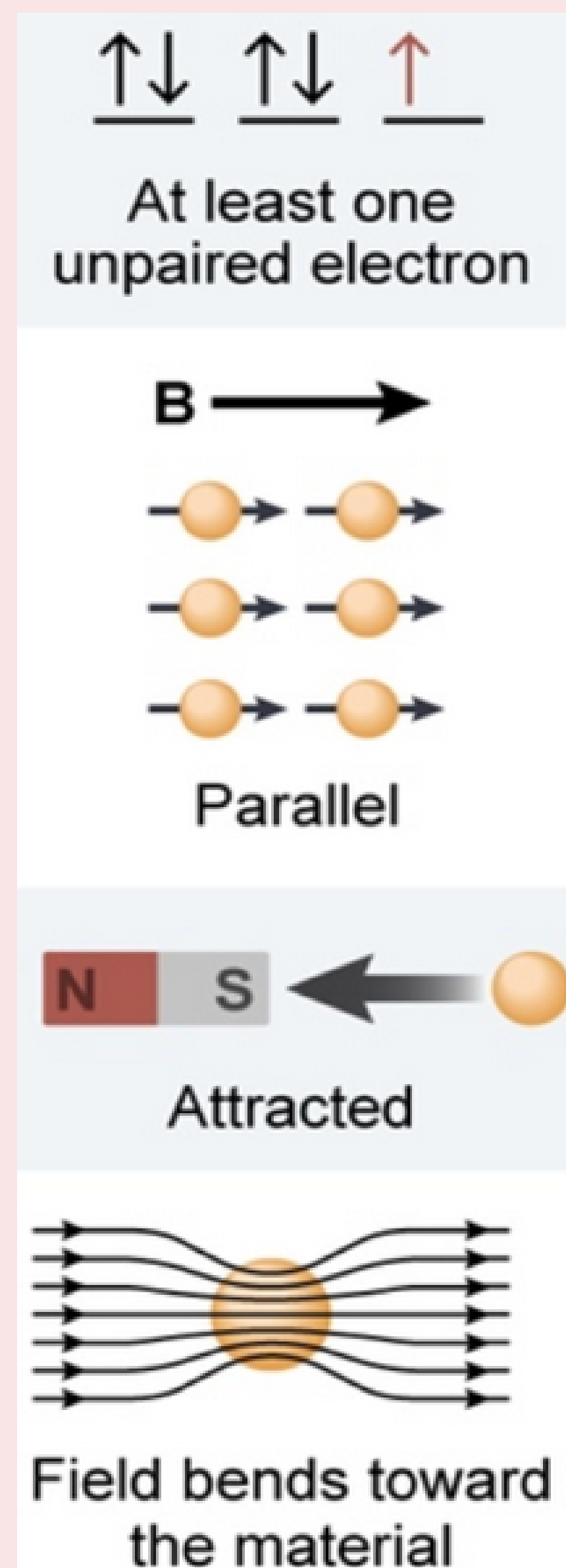


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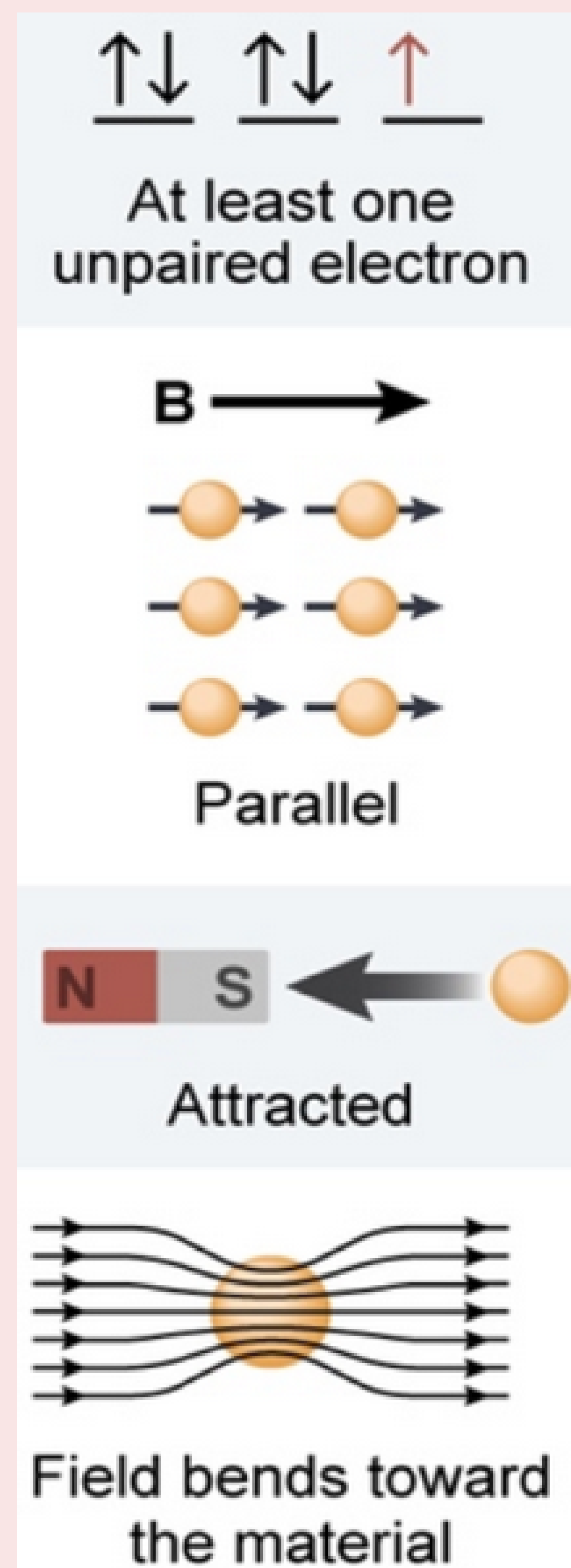


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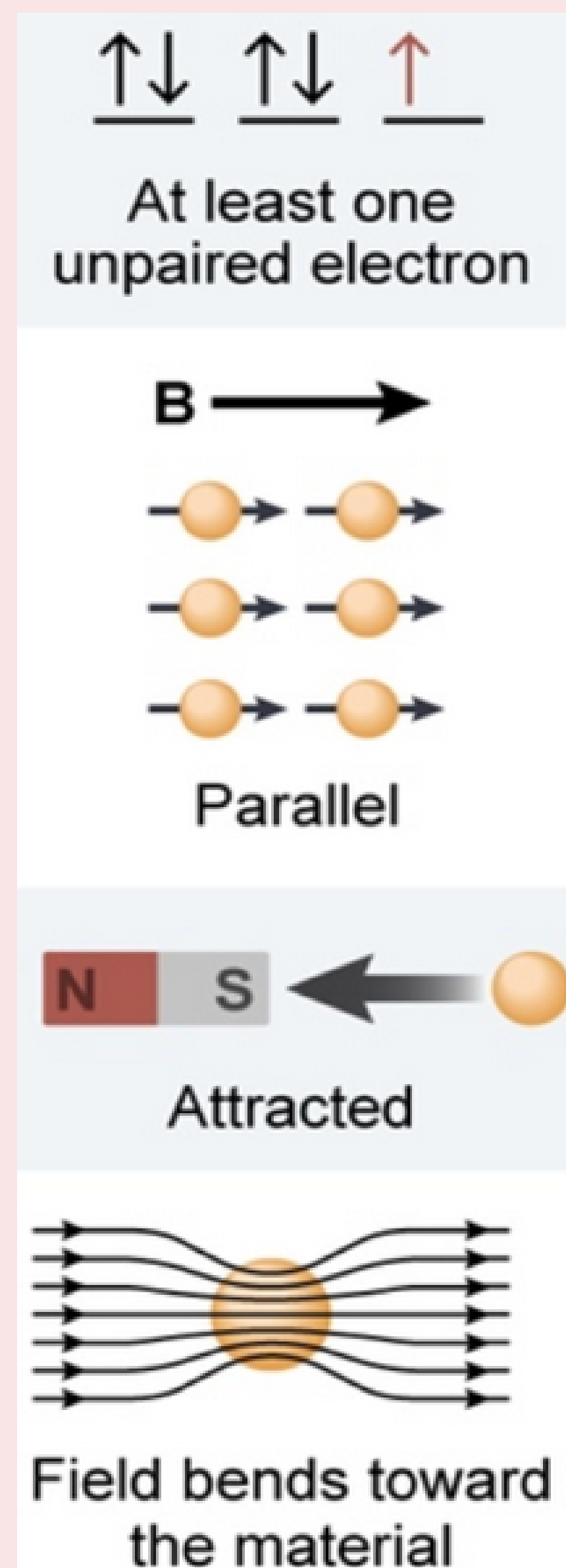


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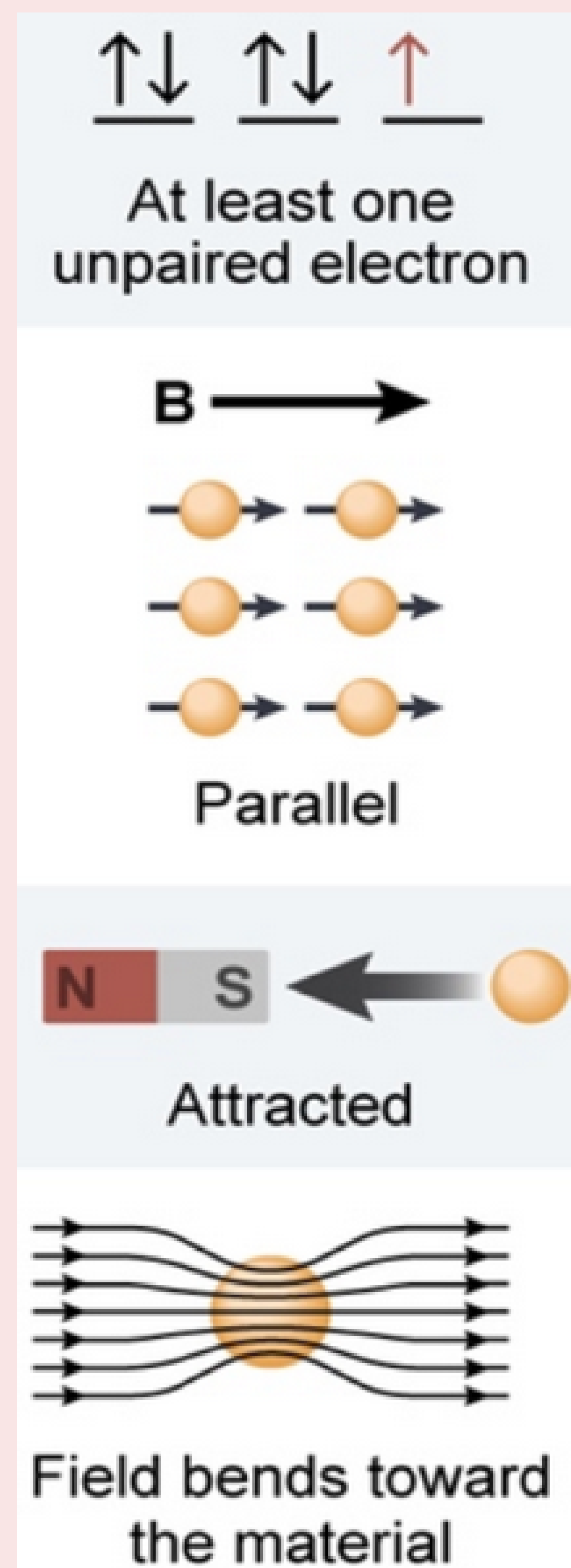


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- Material **does not retain the magnetic properties when the external field is removed**.
- **Susceptibility depends on the temperature** of the material.

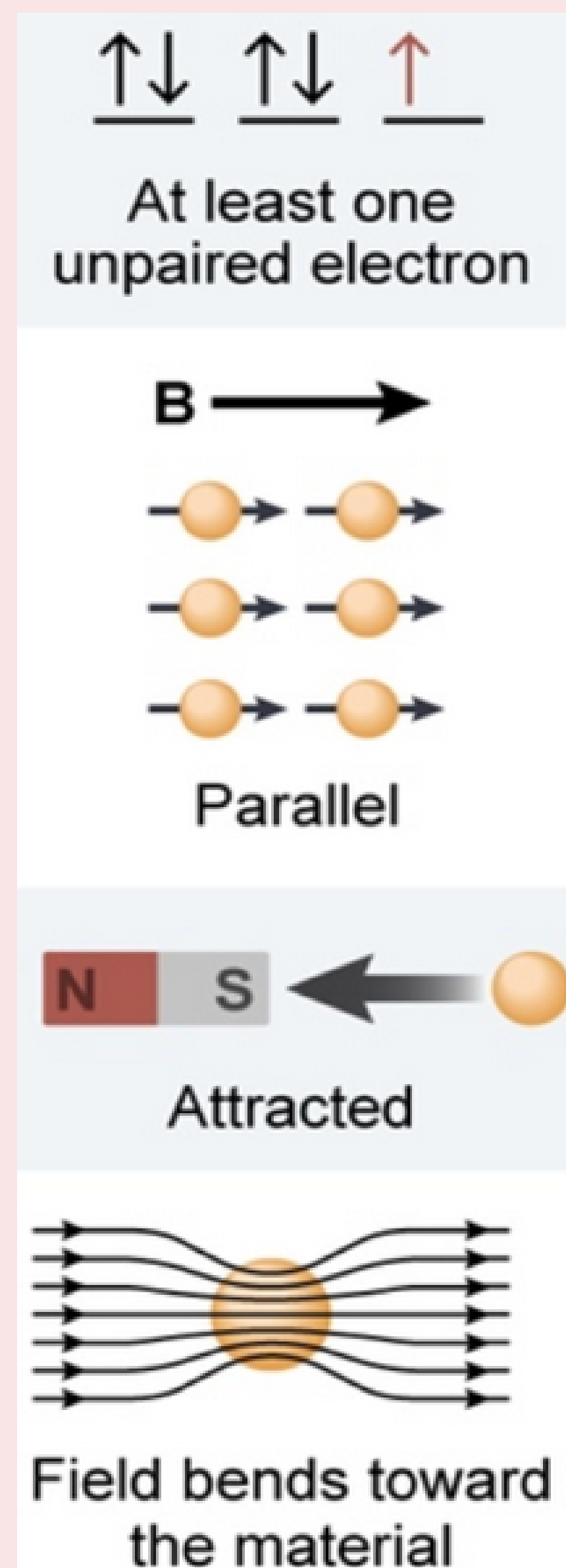


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Paramagnetic materials :

Curie's law :

- The magnetisation of a paramagnetic material is inversely proportional to the absolute temperature T .



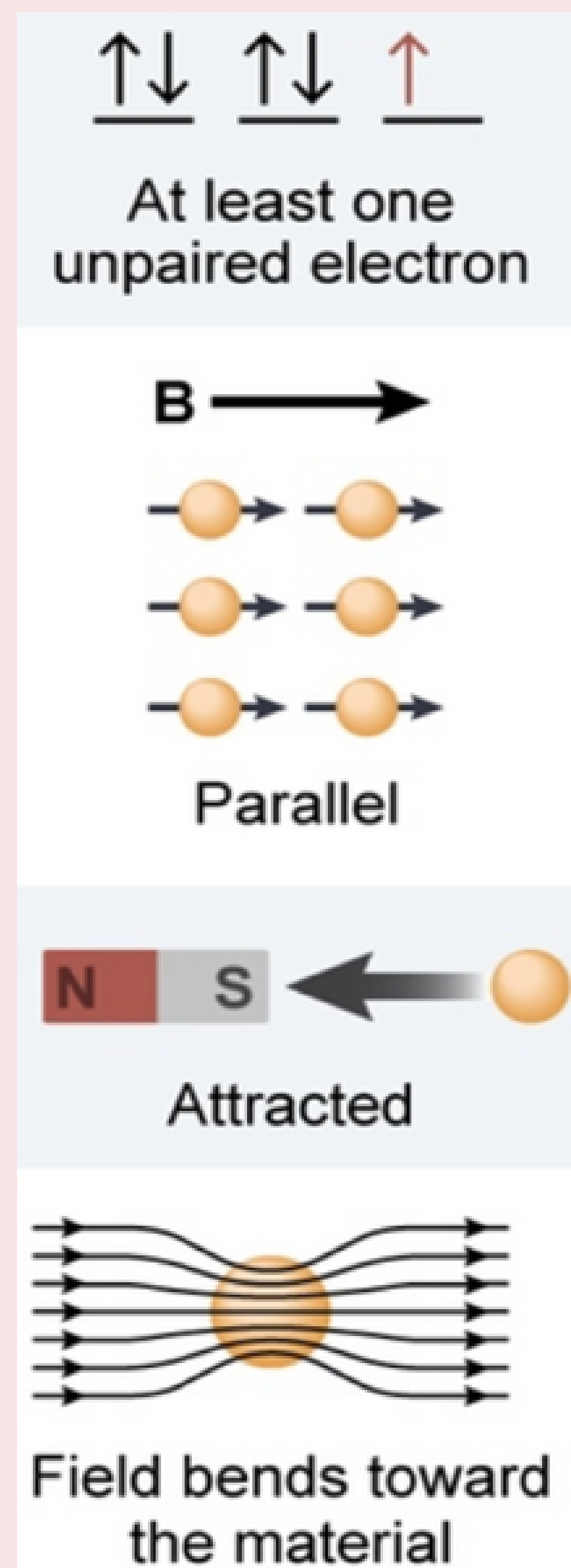
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$$M = C \frac{B_0}{T} \quad \text{Or}$$



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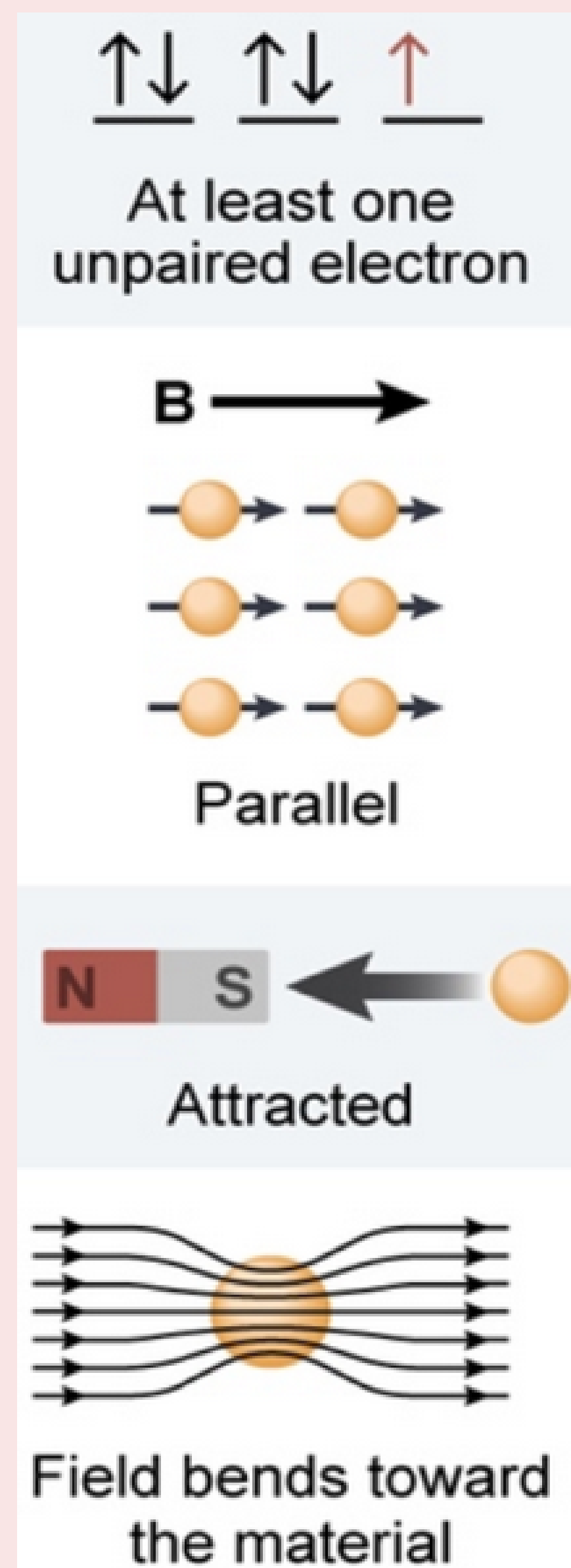
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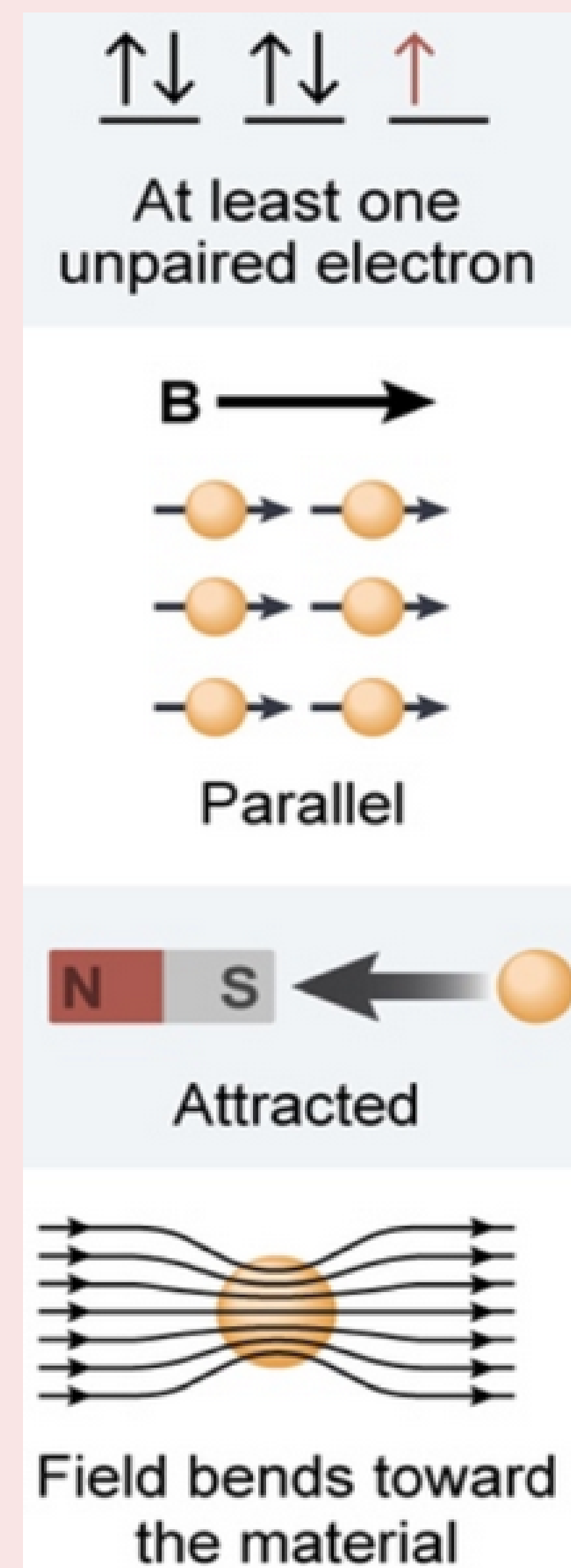
$$M = C \frac{B_0}{T} \quad \text{Or } \chi = C \frac{\mu_0}{T}$$

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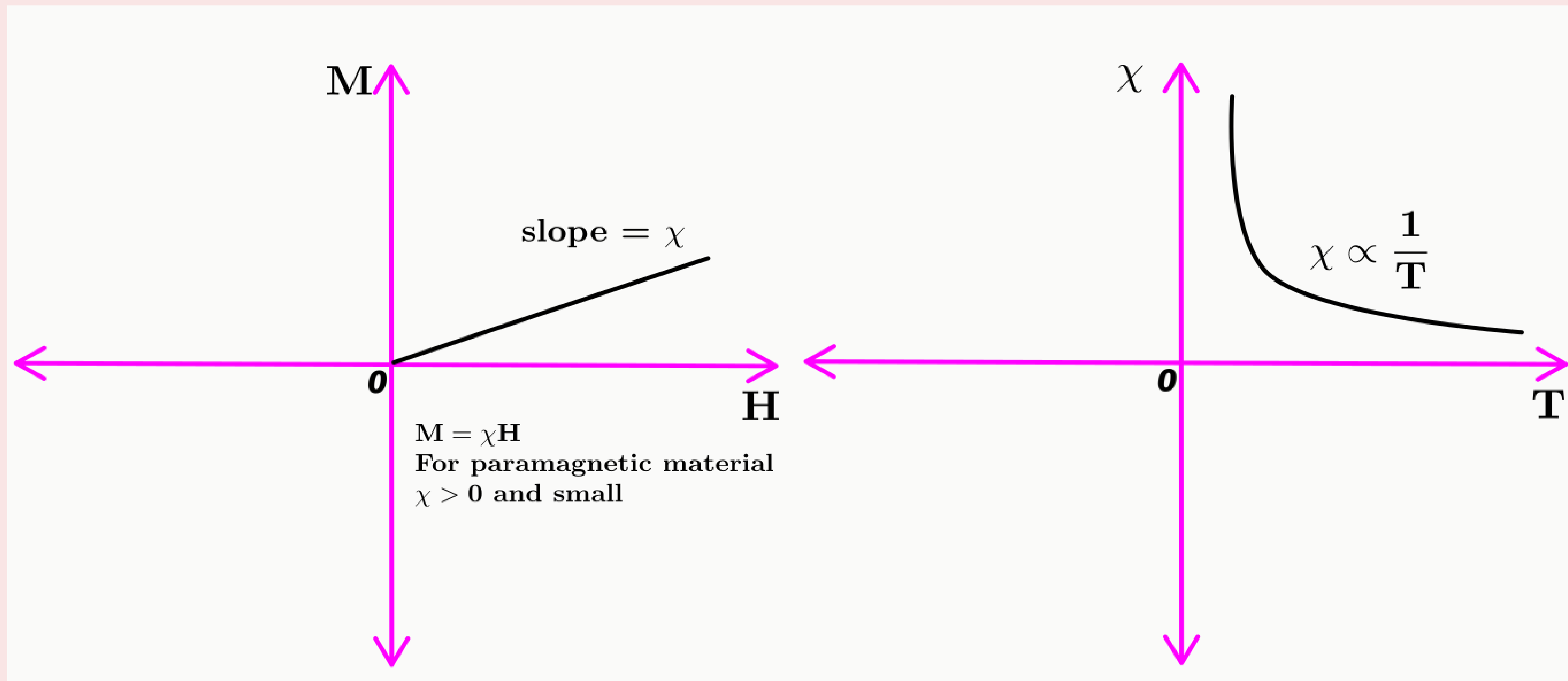
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- The constant C is called Curie's constant.
- Thus, for a paramagnetic material both χ and μ_r depend not only on the material, but also on the sample temperature.

Classification of magnetic materials

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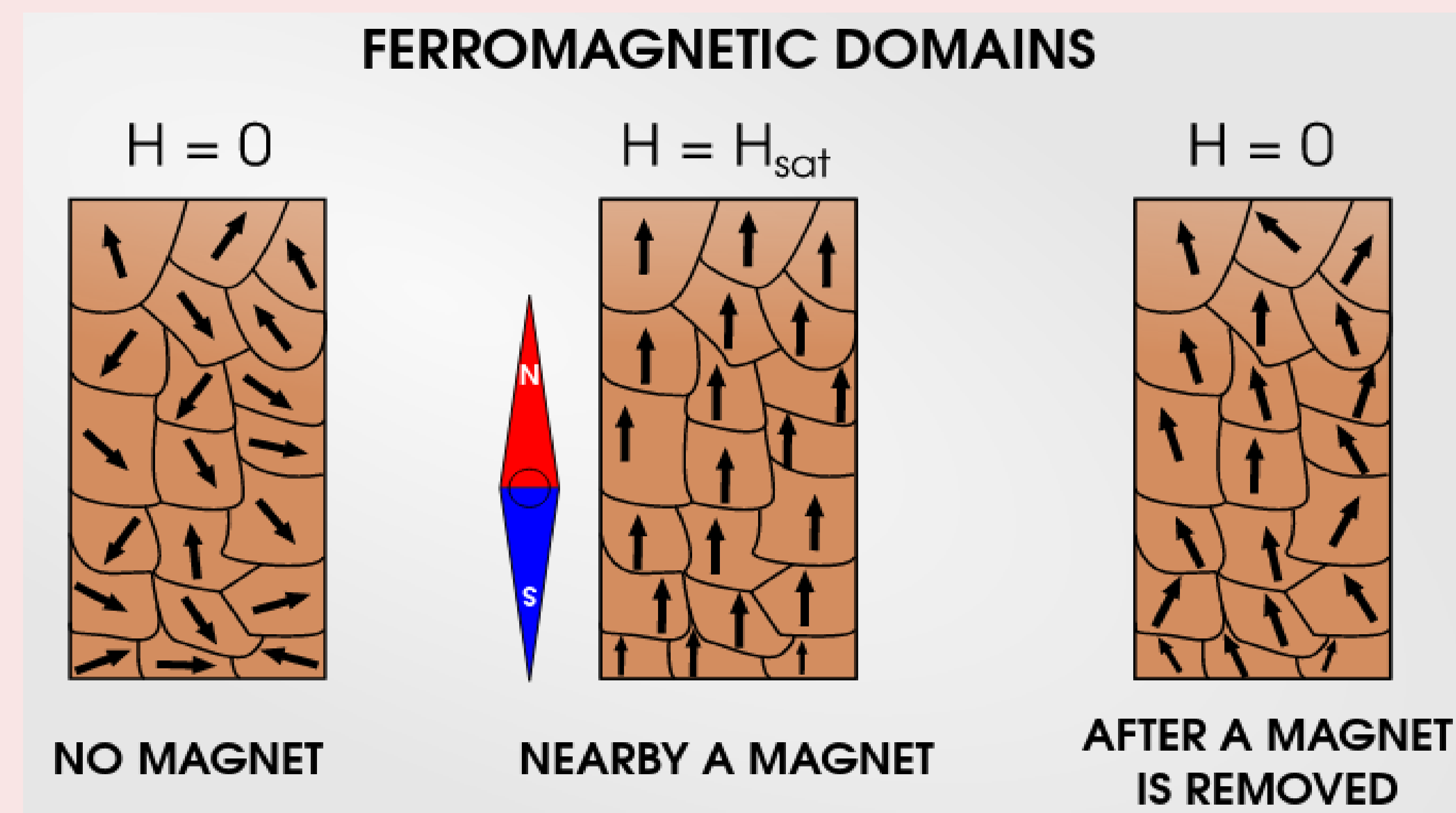
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- **Examples:** Iron, cobalt, nickel



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- **Explanation: Ferromagnetism is a special case of paramagnetism.**

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Ferromagnetic materials :

- **Explanation: Ferromagnetism is a special case of paramagnetism.** In ferromagnetic substances, the contribution of the spin magnetic moment to the **magnetic dipole moment of atoms is very large.** They **possess strong magnetic properties** due to the **presence of magnetic domains.**

Classification of magnetic materials

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- **Explanation:** Ferromagnetism is a special case of paramagnetism. In ferromagnetic substances, the contribution of the spin magnetic moment to the **magnetic dipole moment of atoms is very large**. They **possess strong magnetic properties** due to the **presence of magnetic domains**. In these domains, large number of atomic moments (10^{12} to 10^{15}) are aligned parallel, so that the magnetic force within the domain is strong.

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- **Susceptibility depends on the temperature** of the material

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A ferromagnetic material behaves as a paramagnetic material when temperature is greater than the Curie temperature T_c .

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Paramagnetic materials :

